

Exercise One (1)

(a) Define the term Search-based Software Engineering

(b) What do you understand by the term Meta-heuristics?

(c) What do you understand by the term Optimization?

(d) What are Heuristics Optimization techniques

(e) Differentiate between the following terms as used in meta-heuristics with two in each case

(i) Greedy algorithms versus Optimization Algorithms

(ii) Degree of Exploitation and degree of Exploration

(iii) Single-solution meta-heuristics and evolutionary computing(Population-based methods)

(iv) Genotype and Phenotype

(v) Genetics Algorithm and Tabu Search

(V) Genetics Algorithm and Traditional Algorithms.

Exercise Two (2)

(a) Explain the processes involve in order to develop an optimization algorithm

or

Explain three key ingredients to develop optimization algorithms for search-based Software Engineering in order to solve Software Engineering problems.

Exercise Three (3)

(a) List five variants of Evolutionary Algorithms and hence explain two of them.

Answer:

(1) Genetics Algorithm

- (2) Genetics Programming
- (3) Differential Evolution
- (4) Evolution Strategies
- (5) Evolution Programming

Explanation of two of them:

- (1) Genetics Algorithm
- (2) Genetics Programming

(b) With respect to a generalized annotated Evolutionary Algorithm diagram, explain the concept of Evolutionary Algorithms.

Exercise Four (4)

(a) List two classifications of Meta-heuristics algorithms with three examples in each category and hence with respect to your chosen project, explain one algorithm explicitly in **each category**.

Exercise Five (5)

- (a) Explain the roles of different memories in a Tabu Search
- (b) Given the Pseudocode for a classical Tournament selection scheme below:
 - (i) Explain each line of the Tournament Algorithm
 - (ii) With respect to your chosen project, identify that gaps in the pseudocode below and modify the gaps by introducing a new Tournament Algorithm with comments on each line.
- (c) List five types of mutation search operators used in heuristic optimization and hence explain one.
- (d) List three selection schemes used in heuristic optimization and hence explain two.

(e) Explain two disadvantages of hill climbing algorithm.

(f) Effective search balances Exploitation of the best solution so far and Exploration of the search space. Explain this state explicitly.